Invasive species: ballast water battles

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Invasive species spread through ballast water are now considered to be among the most dangerous environmental hazards facing the planet. This article looks at some of the regulatory measures currently being taken to combat the threat.

As a concept, maintaining one geographical region free from living organisms that are endemic to another region is not new. It has existed in the form of quarantine controls and been applied to all means of transportation.

As far as ships are concerned, as early as the 18th century, the Spanish maritime authorities issued 'Ordenanzas Generales de las Capitanias de Puerto (1783)'. Article 138 read: 'It is forbidden to discharge ballast in water and to ballast and de-ballast without a permit from the harbour master. It is also forbidden to load sand ballast unless it is an absolute emergency. The ballasting and de-ballasting operations should be carried out with the appropriate precautions.'

Turning to Article 142, the ordinance stated that: 'On matters regarding damage to the port by dumping debris and ballast, notwithstanding the application of the fine, it will be necessary to investigate the intention to be able to punish the perpetrators in accordance with the law' (source: Prefectura Naval Argentina, 2009).

This may have been one of the first attempts to regulate ballast discharges.

With the introduction of steel-hulled vessels and the use of water as ballast, the problem of invasive species became even more pertinent due to the larger quantities of ballast transported and, implicitly, the increased number of species moved from one place to another.

The development of larger and faster ships completing their voyages in ever shorter times, combined with rapidly increasing international trade, meant that the natural barriers to the dispersal of species across the oceans were being reduced.

As a result, the spread of invasive species is now recognised as one of the greatest threats to the ecological and economic well being of the planet.

These species are causing enormous damage to bio-diversity. The valuable natural riches of our planet, upon which we depend, are under threat. Direct and indirect health effects are becoming increasingly serious and the damage to nature is often irreversible. Aquatic invasions – considered the second greatest threat to global bio-diversity after habitat loss – are virtually irreversible, and increase in severity over time.

Economic impact

The global economic impact of invasive aquatic species (IAS) have not been thoroughly quantified but are likely to be in the region of tens of billions of US dollars a year or more. Human health impacts can also be caused by the transfer and spread of pathogens and toxic organisms such as harmful algae in ships' ballast water.

In late 2004, the GloBallast Programme – a cooperative initiative by the IMO, the Global Environment Facility and the United Nations Development Programme (UNDP) – undertook an initial scoping study on the global economic impacts of invasive aquatic species. The study identified the current state of knowledge in relation to both direct economic impacts and the costs of responding to IAS.

Direct economic impacts are the actual monetary costs caused by the species in their invaded environments, including costs from reductions in fisheries production, closure/reductions in aquaculture, physical impacts on coastal infrastructure (fouling), reduction in economy of shipping (fouling) and closure of recreational and tourism beaches.

The data relating to seven specific invasions indicated that the direct economic impacts of these seven species alone are more than US $10 billion a year (GloBallast, 2004). It was estimated that the direct economic impact from all the current IAS may be of the order of US $100 billion a year.

The response costs identified by the study were the costs incurred by society in responding to the problem, including prevention, control and eradication, research and monitoring, education and communication, compliance monitoring and enforcement and the effort to develop new ballast water treatment technologies.

Overall, the projected costs of responding globally are estimated at up to around four per cent of the total global economic impact.

The IMO response

The problem is global in scope and cross-sectoral coordination and cooperation are imperative. Preventing the international movement of invasive species and coordinating a timely and effective response to invasions requires cooperation and collaboration among governments, economic sectors, non-governmental and international organisations.

The IMO teamed up to address this issue with the UNDP, Global Environment Facility (GEF), the United Nations Environment Programme (UNEP) and the International Union for Conservation of Nature (IUCN) and stands prepared to increase coordination with other organisations that share the same concerns.

It is, therefore, not surprising that ships' ballast water management has become an issue of the highest priority for a wide group of stakeholders including IMO member states, the shipping industry, environmentalists, technology developers

and technology test facility operators.

During the last two decades, the IMO has been working constantly to address, meet and respond to the challenges associated with ballast water management. Initially this has been through the development of two sets of guidelines and, more recently, by drafting a new legally binding international instrument, the International Convention for the Control and Management of Ships' Ballast Water and Sediments, which was adopted in February 2004.

The Convention aims to prevent, minimise and ultimately eliminate the risks to the environment, human health, property and resources arising from the transfer of harmful aquatic organisms in ships' ballast water.

This Convention will enter into force 12 months after the date on which not less than 30 states, the combined merchant fleets of which constitute not less than 35 per cent of the gross tonnage of the world's merchant shipping have ratified it.

As of February 2010, the Ballast Water Management (BWM) Convention has been ratified by 22 countries representing almost 23 per cent of the world's merchant shipping capacity (for the latest figures see 'Status of Conventions' at http://wwww.imo.org).

Together with its associated Guidelines, the Convention provides the much needed framework for developing an integrated-systems approach to ballast water management. It offers the critically needed set of management tools through which the maritime industry can be regulated in a manner that is practical and transparent.

The Convention is centred on the precautionary approach principle and gives due consideration to the environmental benefits, technological achievability and, most important, to global standardisation.

Like most IMO conventions, this instrument consists of articles and an annex containing the technical regulations. Two appendices provide the format of the BWM certificate and the BW record book. While many of the articles are common to all IMO instruments (such as application, survey and certification, technical assistance, cooperation and regional cooperation, entry into force and amendments), the technical regulations, grouped in five sections, are more specific and provide, among other things, the ballast water management standards and their sequential 'phase in' application.

The ballast water exchange standard, contained in regulation D-1, requires ships to exchange a minimum of 95 per cent ballast water volume in open seas and the ballast water performance standard, contained in regulation D-2, requires that the ballast water discharged meet certain organism concentrations below specified limits.

The Convention differentiates between ships constructed before 2009 and those after this year and includes a 'phase-in' procedure, contained in regulation B-3, which will eventually phase out ballast water exchange by 2019 (see following article).

Treatment technologies

As the performance standard appears to be more stringent and is the preferred option in the long term, regulation D-3 establishes the approval requirements for ballast water management systems (BWMs) to treat the water in order to comply with the D-2 standard.

It is increasingly believed that the use of active substances – substances or organisms that have a general or specific action on or against harmful aquatic organisms and pathogens – will become a condition to comply with the provisions of the Convention.

Active substances accomplish their intended purpose through action on aquatic organisms and pathogens in ships' ballast water and sediments. However, if the ballast water is still toxic at the time of discharge into the environment, the ecosystem in the receiving port may suffer unacceptable harm.

A cautious approach needs to be taken by developers of such ballast water management systems and thorough toxicity testing is needed to determine if an active substance can be used and under which conditions the potential of harming the receiving environment or human health is acceptably low.

Traditionally, compliance monitoring and enforcement of IMO conventions are the prerogatives of the member states and the various functions are performed by the respective administrations or their designated organisations, in many cases, classification societies.

Due to the uncertainty associated with discharging chemically treated ballast water from a ship operating under the authority of one government in some other country's port, compounded by the fact that no obvious impacts can be seen at the time of the discharge, the BWM Convention is perhaps one of the very rare cases when the IMO is called upon to ensure proper application with due observance of the safeguards required by it.

According to regulation D-3.2 of the BWM Convention, ballast water management systems that make use of active substances are approved by the IMO and the procedure for approval of such systems – adopted by the Marine Environment Protection Committee in 2008 by Resolution MEPC.169(57) – provides the safeguard for the sustainable use of active substances.

The procedure consists of a two-tier process – basic approval and final approval – that involves extensive consultations among IMO member states and independent scientific advice from a technical group called to evaluate the physical and chemical hazards and ensure that the respective ballast water management system does not pose an unreasonable risk for environment, human health, property or resources.

Such a technical group was established by the MEPC in 2006 under the auspices and with the direct support of the Joint Group of Experts on the Scientific Aspects of Marine Environmental Protection (GESAMP).

GESAMP

The GESAMP is an advisory and multi-disciplinary body consisting of specialised experts nominated by the group's sponsoring agencies: the IMO, Food and Agriculture Organisation (FAO), the Intergovernmental Oceanographic Commission of the UN Education, Scientific and Cultural Organisation (UNESCO-IOC), the World Meteorological Organisation (WMO), the World Health Organisation (WHO), the International Atomic Energy Agency (IAEA), the UN, UNEP and the UN Industrial Development Organisation.

Experts working for the GESAMP Ballast Water Working Group (GESAMP–BWWG) act independently in their individual capacity and are bound by a statement of acceptance to ensure that proprietary data is treated as confidential.

The manufacturers of the ballast water management systems should evaluate the active substance and the potential discharge of treated ballast water in accordance with the approval criteria specified in the procedure.

Upon completion, a manufacturer should prepare an application dossier on the respective BWMs that makes use of active substances and submit it to the administration of the member of the organisation concerned. The administration should evaluate the application dossier and confirm that it is satisfactory and complete before submitting a proposal

for approval to IMO.

It is particularly important that a non-confidential description of the respective system is circulated well in advance to allow members of the organisation, the UN and its specialised agencies, inter-governmental organisations having agreements with the IMO and non-governmental organisations in consultative status with the IMO, to submit information that may be relevant to the evaluation.

After the two-tier process is completed and final approval has been granted by the Marine Environment Protection Committee of IMO, administrations, shipowners and other stakeholders can be sure that the system is robust, safe and environmentally friendly and the proponent administration can issue the type approval certificate.

If, in the past, the lack of technologies and management options were to be a matter of concern for those IMO members contemplating the ratification of the BWM Convention, the recent surge in the development of groundbreaking ballast water treatment technologies has removed this last barrier in the implementation process. The shipping industry and the international community as a whole need to recognise that ballast water management options are currently available.

Out of eight ballast water management systems that have received final approval from IMO, five have been certified for type approval by their national administrations and are readily available for use on board ships, while 17 more systems are currently under evaluation by the Organisation.

The momentum precipitated by technology developers needs to be sustained and the wide ratification of the BWM Convention will provide the necessary guarantee that their efforts will be rewarded by the shipping industry by acting decisively to address the issue of aquatic invasive species in ships' ballast water.

Emerging solutions

The BWM Convention is a far-reaching instrument and remains open to new ideas. Regulation B-3 allows for alternatives to the ballast water exchange and treatment, provided that such new methods ensure at least the same level of protection to the environment, human health, property or resources and are approved in principle by the IMO. This calls for further R&D initiatives and opens the door to engineering ingenuity.

Several attempts have been made by various member states to define the concept of 'same level of protection' and to establish some sort of equivalency between various solutions. The matter is currently on the agenda of one of the IMO technical sub-committees and GloBallast partnerships, through its Global Industry Alliance (GIA), has once again offered to assist by organising an expert workshop to establish a scientific basis for equivalency between treatment technologies approved under guidelines (G8) and procedure (G9) and alternative management options.

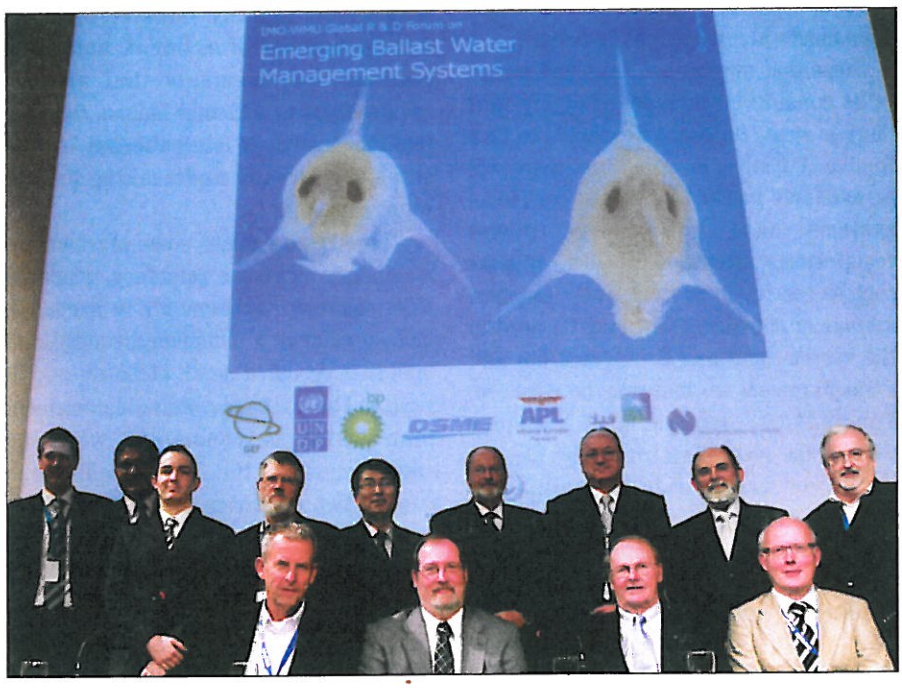
The workshop was part of a series of events, including an R&D Forum, see picture below, organised by GloBallast in cooperation with the World Maritime University in January 2010. The Forum, held from 27 -29 January 2010 in Malmö, Sweden, was attended by some of the world's largest maritime organisations involved in shipping, ship building and design, maritime R&D institutes, technology developers, environmental organisations and academia.

Part of an older tradition initiated by GloBallast in 2002, it provided a platform to share the latest research in the field, and to facilitate an open debate and a review of the existing and emerging ballast water management systems. The Forum represented a step forward in the effort to outline the strategy to meet the current challenges and to eliminate the risks posed by invasive species transferred by ships.

The IMO Secretary General has emphasised on several occasions that the effectiveness and efficiency of maritime transport and, by extension, the IMO's work, can have a major and direct impact on the global economy and environmental sustainability. The organisation, therefore, remains, committed to reducing the negative impact of shipping's daily operations on the environment.

Further confirmation of the importance attached to aquatic invasions comes from the Marine Environment Protection Committee which has initiated the development of international measures for minimising the transfer of invasive species through bio fouling of ships.

IMO will continue, through the Integrated Technical Cooperation Programme, to assist its members in their efforts towards implementation of effective measures to address aquatic invasions, however, all these efforts are futile without the most important ingredient, the determination of member states. The international community needs vision, foresight, purpose and strength of will. All the stakeholders need to act now, pro-actively, positively and with due sense of responsibility in preserving our planet for the future generations.



The Forum was an important opportunity for truly international discussion of the whole range of ballast water treatment technology. Over 30 treatment systems are under development around the world, and this was a unique opportunity for real comparisons to be made. The ballast water convention will come into force when 30 nations and 35 per cent of the world's tonnage have ratified it; already 22 nations, representing 23 per cent of the world's tonnage, have ratified. This means that the Forum came at a very useful point for the global shipping industry, which is benefiting from the Forum's examination of the technologies available. Formal conclusions and advice from the Forum are now being prepared, and will be submitted to IMO's committees in the coming months.
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